

B1GMB42 Software Manual

IndianaDell workspace

2026-07-09

B1GMB42 Software Manual

Machine: Dell Precision Tower 5810 (B1GMB42)

Hostname: Tower5810

OS: Ubuntu 26.04 LTS (resolute)

Workspace: ~/Documents/IndianaDell

Companion hardware manual: B1GMB42-slot-port-inventory.md (slots, GPUs, storage, PERC, ports)

This manual documents every **host-facing install** the IndianaDell workspace provides: apt packages, rustup, Python venvs, built tools, Flatpak apps, GNOME preferences, Plymouth themes, optional GPU/ROCm tooling, ZFS recovery, Ventoy live persistence, and GitHub sync. Each chapter covers one topic using the same structure:

1. What gets installed
2. How it is installed
3. How to verify
4. How to customize
5. What bin/rebuild-machine does and does not do

Build PDFs: bin/build-all-docs (all manuals) or bin/build-software-manual (this book only).

Quick reference: docs/features-available.md (cheat sheet, not a replacement for this manual).

GitHub: <https://github.com/webaugur/IndianaDell> (private)

Supersedes: flat B1GMB42-software-inventory.md (now a stub with links here).

Chapter 1 — Introduction

What this workspace installs

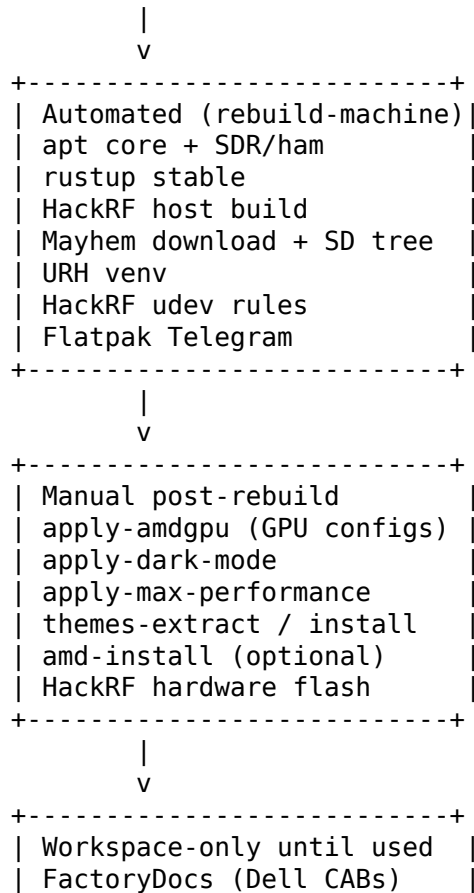
IndianaDell is a **software restoration toolkit** for Tower5810 after a fresh Ubuntu 26.04 install. It does not partition disks, configure ZFS, flash BIOS, or install Windows. It restores the development, SDR, ham radio, HackRF/Mayhem, documentation, and workstation utility stack documented in the chapters that follow.

The workspace lives at `~/Documents/IndianaDell`. Copy or clone it before running `bin/rebuild-machine`.

Install layering

Software arrives in three layers. Understanding the order prevents skipped steps after a reinstall.

Fresh Ubuntu 26.04





Automated steps run via `bin/rebuild-machine` (see Chapter 2). **Manual** steps are intentional: GPU session files, GNOME gsettings, Plymouth overlay, and hardware flashing need user context or `sudo` at the right time. **Workspace-only** content stays in the repo until you invoke the matching `bin/launcher`.

Workspace vs host paths

Location	Role
<code>~/Documents/IndianaDell/</code>	Source of truth for scripts, themes, HackRF assets
<code>/usr/</code>	Apt-installed binaries, Plymouth themes, udev rules (after apply)
<code>~/ .cargo/</code>	Rust toolchain (rustup)
<code>hackrf/venv-urh/</code>	Universal Radio Hacker Python venv
<code>hackrf/build/</code>	HackRF host tools built from source
<code>hackrf/local/</code>	Optional CMAKE_INSTALL_PREFIX for built libhackrf
<code>Themes/*/mirror/</code>	Frozen copies of apt-owned theme files
<code>FactoryDocs/</code>	Dell vendor packages (not auto-installed to host)

Reading guide

If you need...	Read
Full restore after reinstall	Ch. 2 + Ch. 3
Python, Rust, pandoc	Ch. 4
Boot/login/desktop look	Ch. 5 + Ch. 7
FirePro GPUs, ROCm	Ch. 6
GNU Radio, gqrx, SoapySDR	Ch. 8

If you need...	Read
fldigi, WSJT-X, CHIRP	Ch. 9
HackRF, Mayhem, URH	Ch. 10
Telegram	Ch. 11
iotest, dellmerge	Ch. 12
Dell driver CABs	Ch. 13
Known gaps	Ch. 14
Ventoy live persistence, Grok autostart	Ch. 15
ZFS rpool / bpool recovery	Ch. 2 + docs/B1GMB42- zfs-recovery.md (+ Ch. 15 live boot)
/etc/default/zfs force import	Ch. 2 / ZFS recovery manual — ZPOOL_IMPORT_OPTS="- f"
All bin/ commands	Appendix A
All apt package names	Appendix B

PATH and launchers

IndianaDell bin/ and scripts/ directories are prepended to PATH via `~/ .config/indianadell/path.sh` (sourced from `~/ .bashrc`). Project tools override same-named system binaries.

Related documents

- **Hardware:** B1GMB42-slot-port-inventory.md + PDF — GPUs, PERC, bays, ports
- **ZFS recovery:** docs/B1GMB42-zfs-recovery.md + PDF — live-media rpool/bpool chroot
- **PERC IT flash:** docs/B1GMB42-perc-it-flash.md — H710 Free-DOS/Wiggly path
- **Themes deep-dive:** Themes/README.md and per-folder READMEs
- **HackRF inventory:** hackrf/MANIFEST.txt
- **Apt snapshots:** apt-full-manifest.txt, apt-hamradio-dev-manifest.txt
- **Rebuild log:** scripts/rebuild/last-run.log

Chapter 2 — Rebuild and Recovery

What gets installed

bin/rebuild-machine restores the full automated software stack in one run (~15–30 minutes, network dependent). It installs **91 apt packages** from scripts/rebuild/package-lists.sh (APT_CORE 38 + APT_SDR_HAM 53), plus rustup, HackRF repos/build, Mayhem v2.4.0 assets, URH venv, udev rules, and Flatpak Telegram.

How it is installed

```
cd ~/Documents/IndianaDell
chmod +x bin/* scripts/rebuild/*.sh
bin/rebuild-machine           # full restore
bin/rebuild-machine --verify-only # check only, no installs
```

Environment overrides:

Variable	Effect
SKIP_TELEGRAM=1	Skip Flatpak Telegram install
SKIP_HACKRF_BUILD=1	Skip cmake build of HackRF host tools

Phases (from scripts/rebuild/rebuild-machine.sh):

Phase	Action
1	apt-get update
2	Install APT_CORE (38 packages) — build, Python, docs, GPU utils, flatpak, gh
3	Install APT_SDR_HAM (53 packages) — GNU Radio, ham, SDR hardware, HackRF
4	Flatpak remote + org.telegram.desktop (unless skipped)
5	rustup stable if rustc missing
6	Clone HackRF/Mayhem/URH repos under hackrf/repos/
7	Build HackRF host tools to hackrf/build/ (unless skipped)
8	Download Mayhem v2.4.0 + extract SD card tree
9	Create hackrf/venv-urh/ with URH

Phase	Action
10	Install HackRF udev rules; chmod bin/ and scripts
11	Regenerate apt manifests; run verify_stack

Debconf preseed: xastir/install-setuid is set to false before apt to avoid interactive hangs.

Log file: scripts/rebuild/last-run.log

How to verify

bin/rebuild-machine --verify-only

verify_stack checks:

- Every package in APT_CORE and APT_SDR_HAM via dpkg-query
- Commands: rustc, cargo, gnuradio-config-info, grcc, gqrx, fldigi, wsjtx, chirpw, hackrf_info, inspectrum, pandoc, xelatex, vkcube
- hackrf/venv-urh/bin/urh
- Mayhem firmware zip and extracted SD tree
- Built hackrf/build/hackrf-tools/src/hackrf_sweep
- Launchers: dellmerge, gpu-stress, iotest, apply-amdgpu, rebuild-machine
- Flatpak Telegram (unless SKIP_TELEGRAM=1)

Exit code 0 means all checks passed.

How to customize

- **Add apt packages:** Edit scripts/rebuild/package-lists.sh, update Appendix B, re-run rebuild.
- **Pin HackRF/Mayhem version:** Edit hackrf/scripts/download-mayhem.sh and MANIFEST; rebuild does not auto-upgrade pinned releases.
- **Skip heavy steps:** Use SKIP_* env vars for CI or partial recovery.

What rebuild does / does not do

Rebuild does	Rebuild does not
apt install all listed packages rustup, HackRF build, Mayhem download	Partition disks or ZFS sudo bin/apply-amdgpu

Rebuild does	Rebuild does not
URH venv, udev rules	bin/apply-dark-mode / apply-max-performance Plymouth theme install
Regenerate apt-full-manifest.txt chmod workspace scripts	Flash HackRF / PortaPack firmware bin/amd-install (ROCm) Install FactoryDocs CABs to Windows

After a successful rebuild, continue with **Chapter 3 — Post-Rebuild Checklist**.

ZFS rpool / bpool recovery

Full guide: docs/B1GMB42-zfs-recovery.md + B1GMB42-zfs-recovery.pdf (also on **DOSBOOT** IndianaDell/recovery/).

Required on the installed system: /etc/default/zfs must set force import so boot can recover after export or unclean shutdown:

```
# /etc/default/zfs
ZPOOL_IMPORT_OPTS="-f"
```

Verify anytime: `grep ZPOOL_IMPORT_OPTS /etc/default/zfs`. One-shot boot alternative: kernel cmdline `zfsforce=1`.

Quick (with scripts, from Ventoy live):

```
sudo apt-get install -y zfsutils-linux
cd ~/Documents/IndianaDell # or DOSBOOT/IndianaDell/recovery
sudo ./mount-rpool-recovery.sh mount
sudo ./scripts/recovery/mount-bpool-recovery.sh mount
sudo ./mount-rpool-recovery.sh chroot
# inside chroot: confirm ZPOOL_IMPORT_OPTS="-f", repair, update-initramfs/grub
```

Without scripts: manual `zpool import -N -f -R /recovery rpool`
— see ZFS recovery manual Section 3.

Build PDF: `bin/build-zfs-recovery-doc`. Deploy to DOSBOOT:
`bin/deploy-dosboot-recovery`.

See also **Chapter 15** for Ventoy live boot.

Chapter 3 — Post-Rebuild Checklist

`bin/rebuild-machine` intentionally stops before steps that need a logged-in desktop, a reboot, or hardware attached. Run this checklist once per fresh install.

1. GPU session configuration

Tower5810 has three AMD FirePro cards (W5000/W5100). Multi-GPU Wayland/X11 configs live in etc/.

```
cd ~/Documents/IndianaDell
sudo bin/apply-amdgpu
sudo reboot
```

Verify after reboot: echo \$WAYLAND_DISPLAY, glxinfo -B, vkcube on each display if needed.

See Chapter 6 for ROCm (bin/amd-install) — optional and not supported for ML on these GPUs.

2. GNOME session preferences

Run as the **desktop user** (not root):

```
bin/apply-dark-mode           # Yaru-dark GTK, shell, icons, GDM greeter
bin/apply-max-performance    # no suspend, dimming, or night light
```

Verify:

```
gsettings get org.gnome.desktop.interface color-scheme
powerprofilesctl get
```

See Chapter 7 for every gsettings key touched.

3. Boot splash (optional)

Default Ubuntu **bgrt** Plymouth theme shows Dell BGRT center + Ubuntu watermark. To customize:

```
bin/themes-extract           # refresh mirrors + extract logos
# edit Themes/boot/overlay/watermark.png or background.png
sudo bin/themes-install-boot  # or --oem / --no-watermark
sudo reboot
```

Restore factory: sudo bin/themes-restore-boot

See Chapter 5.

4. HackRF hardware (when device is available)

```
source bin/hackrf-env
hackrf_info                 # should list board
bin/hackrf-flash-mayhem     # extract USB flash bundle
# follow bundle README for DFU flash
bin/hackrf-prepare-sdcard   # ensure SD tree is extracted
# copy hackrf/sd-card/mayhem-v2.4.0/* to FAT32 microSD root
```

Verify: hackrf_info, on-device Mayhem version, SD apps visible on PortaPack.

See Chapter 10.

5. Documentation PDFs

```
bin/build-all-docs                # software manual + hardware + inventory PDFs
# or:
bin/build-software-manual          # this manual only
```

Outputs: B1GMB42-software-manual.pdf, B1GMB42-slot-port-inventory.pdf, B1GMB42-software-inventory.pdf.

6. Machine inventory baseline

```
bin/dellmerge > blgmb42.report
sudo bin/iotest                    # optional storage survey
bin/gpu-stress 60 vkcube           # optional GPU smoke test
```

7. FactoryDocs recovery (optional)

Only 19 of 101 pre-crash Dell packages are on disk. Re-download per FactoryDocs/README.md and MANIFEST-pre-crash.txt. These are **workspace archives**, not installed by rebuild.

See Chapter 13.

8. ZFS force import (required on this host)

After any reinstall or recovery chroot, confirm the installed system will force-import pools at boot:

```
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs
# must show: ZPOOL_IMPORT_OPTS="-f"
```

If missing, set it in /etc/default/zfs, then `sudo update-initramfs -c -k all`. Without this, boot can hang after a recovery export or unclean shutdown. See Chapter 2 and docs/B1GMB42-zfs-recovery.md.

Quick verification block

```
cd ~/Documents/IndianaDell
bin/rebuild-machine --verify-only
grep '^ZPOOL_IMPORT_OPTS' /etc/default/zfs    # expect "-f"
source bin/hackrf-env
. ~/.cargo/env && rustc --version
```

```
gnuradio-config-info --version
bin/urh --version
```

Summary table

Step	Command	Reboot?
GPU configs	sudo bin/apply-amdgpu	Yes
Dark mode	bin/apply-dark-mode	No
Max performance	bin/apply-max-performance	No
Custom boot	sudo bin/themes-install-boot	Yes
HackRF flash	bin/hackrf-flash-mayhem + DFU	Maybe
ROCm (optional)	bin/amd-install	Yes
All doc PDFs	bin/build-all-docs	No
ZFS force import	check /etc/default/zfs □ ZPOOL_IMPORT_OPTS="-f"	If initramfs updated
Ventoy persistence seed	~/bin/seed-ventoy-persistence.sh	No

Chapter 4 — Development Toolchain

What gets installed

Component	Version / path	Install source
Python pip, venv	3.14 (system) apt	Ubuntu base + apt python3-pip, python3-venv science stack
numpy, scipy, matplotlib PyQt5	apt apt	GNU Radio Companion, some tools
Rust C/C++ build	1.96.1 stable apt	rustup to ~/.cargo/bin build-essential, cmake, pkg-config
Clang/LLVM	apt	bindgen-style Rust, native tooling

Component	Version / path	Install source
SSL/USB/FFTW/Volk apt dev libs ARM cross-compile	apt	SDR and native builds PortaPack Mayhem firmware (gcc-arm-none-eabi)
pandoc + XeLaTeX	apt	Manual PDF generation
Git, curl, wget GitHub CLI (gh)	apt 2.46 (apt)	repos and downloads HTTPS/token fallback; gh auth login for API access

Python bindings verified on this host: gnuradio, SoapySDR, Hamlib (capital H in Python).

Project venv: URH lives in hackrf/venv-urh/ (Chapter 10), not system-wide.

How it is installed

Apt (rebuild Phase 2): packages listed in Appendix B under “Development” and shared dev libraries.

Rust (rebuild Phase 5):

```
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh -s -- -y --default-source ~/.cargo/env
```

Rebuild skips rustup if rustc is already on PATH.

How to verify

```
python3 --version
python3 -c "import numpy, scipy, matplotlib; print('OK')"
. ~/.cargo/env && rustc --version && cargo --version
cmake --version | head -1
arm-none-eabi-gcc --version | head -1
pandoc --version | head -1
xelatex --version | head -1
```

How to customize

- **New Python project venv:** `python3 -m venv myproject/.venv`
&& `. myproject/.venv/bin/activate`
- **Rust toolchain:** `rustup toolchain install stable, rustup component add clippy`

- **Per-project Rust SDR crates:** `cargo add rtlsdr` etc. — no workspace scaffold ships with IndianaDell
- **Manual PDFs:** `bin/build-all-docs` (all docs) or `bin/build-software-manual` (software manual only)

What rebuild does / does not do

Does	Does not
Install all dev apt packages	Create application-specific venvs beyond URH
Install rustup if missing	Pin a non-stable Rust toolchain
<code>chmod</code> workspace scripts	Install IDE or editor plugins

Chapter 5 — Themes (Boot, Login, Desktop)

What gets installed

The **Themes/** module (~194 MB with mirrors) documents and customizes three visual stages:

Stage	What you see	Apt packages	Workspace folder
Boot	Dell BGRT center + spinner + Ubuntu watermark	<code>plymouth</code> , <code>plymouth-theme-spinner</code> , ...	<code>Themes/boot/</code>
Login	GDM on GNOME Shell	<code>gdm3</code> , <code>gnome-shell</code> , <code>yaru-theme-gnome-shell</code>	<code>Themes/login/</code>
Desktop	Yaru GTK, icons, shell	<code>yaru-theme-gtk</code> , <code>yaru-theme-icon</code> , ...	<code>Themes/desktop/</code>

Active Plymouth theme: `bgrrt at /usr/share/plymouth/themes/bgrrt/bgrrt.plymouth`

Extracted boot logos:

- `Themes/boot/extracted/bgrrt-firmware-oem.png` — Dell from UEFI BGRT (`/sys/firmware/acpi/bgrrt/image`)
- `Themes/boot/extracted/ubuntu-watermark-dark.png` — bottom Ubuntu text

Custom Plymouth install target: indianadell theme under /usr/share/plymouth/themes/indianadell

Each subfolder has its own README.md and apt-packages.txt. See Themes/MANIFEST.txt for a one-page map.

How it is installed

Themes are **not** applied by bin/rebuild-machine. Use launchers:

```
bin/themes-extract           # snapshot apt-owned files + extract logos (~193 MB)
sudo bin/themes-install-boot # install custom Plymouth from boot/overlay/
sudo bin/themes-restore-boot # revert to stock bgrt
bin/apply-dark-mode          # login + desktop dark (Chapter 7)
```

Boot overlay workflow:

```
cp my-logo.png Themes/boot/overlay/watermark.png # bottom Ubuntu text only
sudo bin/themes-install-boot
```

```
cp my-splash.png Themes/boot/overlay/background.png
sudo bin/themes-install-boot --oem # replace center (disable BGR)
```

```
sudo bin/themes-install-boot --no-watermark # hide bottom logo
```

Every Plymouth install runs update-initramfs -u — **reboot required** to preview.

How to verify

```
plymouth-set-default-theme -l
plymouth-set-default-theme # shows active theme
ls Themes/boot/extracted/
gsettings get org.gnome.desktop.interface gtk-theme # after apply-dark-mode
```

How to customize

Deep documentation lives in module READMEs — this chapter summarizes:

- **Boot:** Themes/boot/README.md — overlay/, stock/, indianadell/ plymouth definition
- **Login:** Themes/login/README.md — GDM greeter assets mirrored from apt
- **Desktop:** Themes/desktop/README.md — GTK/icon/shell Yaru variants

Center Dell logo comes from BIOS BGRT firmware, not an Ubuntu file. Change it in BIOS setup or use --oem with your PNG.

What rebuild does / does not do

Does	Does not
Install plymouth, gdm3, Yaru via apt (Phase 2-3)	Run themes-extract or themes-install-boot Set dark mode or GDM greeter prefs Copy overlay PNGs into initramfs

Run Chapter 3 checklist items 2 and 3 after rebuild.

Chapter 6 — GPU and Display

What gets installed

Component	Source	Purpose
vulkan-tools, mesa-utils, clinfo etc/ multi-GPU configs	apt (APT_CORE) workspace	Vulkan/OpenGL/OpenCL diagnostics Wayland, X11, udev, GDM tweaks
amd-radeon/ scripts	workspace	Optional ROCm driver install
bin/gpu-stress	workspace	3-GPU Vulkan smoke test

Hardware (this machine): 2x AMD FirePro W5000 + 1x FirePro W5100. Vulkan and OpenCL work for graphics/compute smoke tests. **ROCm ML/HIP is not supported** on these cards (see Chapter 14).

How it is installed

Apt (automated): GPU utility packages install during rebuild Phase 2.

Session configs (manual):

```
sudo bin/apply-amdgpu    # runs etc/apply.sh
sudo reboot
```

etc/apply.sh installs:

- etc/environment.d/99-amdgpu-wayland.conf
- etc/X11/xorg.conf.d/20-amdgpu-multi-gpu.conf
- etc/modprobe.d/amdgpu-multigpu.conf
- etc/udev/rules.d/99-amdgpu-multigpu.rules

- `etc/profile.d/amdgpu-multigpu.sh`
- `etc/gdm3/custom.conf` (if present)

Optional ROCm:

```
bin/amd-preflight      # check prerequisites
bin/amd-install        # full driver stack from amd-radeon/
bin/amd-verify
bin/amd-uninstall      # remove if needed
```

How to verify

```
vkcube                # Vulkan cube (per display)
clinfo | head -30      # OpenCL platforms/devices
glxinfo -B             # OpenGL renderer
bin/gpu-stress 60 vkcube # stress all GPUs ~60s
lspci -nn | grep -i vga
```

How to customize

- Edit files under `etc/` before re-running `sudo bin/apply-amdgpu`
- ROCm install scripts and README in `amd-radeon/` — machine-specific; read preflight output
- Hardware details: `B1GMB42-slot-port-inventory.md` Video section

What rebuild does / does not do

Does	Does not
Install mesa-utils, vulkan-tools, clinfo	Run <code>apply-amdgpu</code>
Ensure <code>bin/gpu-stress</code> is executable	Install ROCm
	Configure monitor layout (use GNOME Settings)

Required post-rebuild: `sudo bin/apply-amdgpu + reboot` (Chapter 3).

Chapter 7 — GNOME Session

What gets installed

GNOME desktop preferences applied via `gsettings` (user session) and optionally GDM (login greeter). No apt packages beyond what Ubuntu desktop already provides (`gdm3`, `gnome-shell`, Yaru themes).

Scripts:

- `scripts/gnome/apply-dark-mode.sh` via `bin/apply-dark-mode`
- `scripts/gnome/apply-max-performance.sh` via `bin/apply-max-performance`

How it is installed

Run as the **logged-in desktop user**, not root:

`bin/apply-dark-mode`

`bin/apply-max-performance`

apply-dark-mode

Sets:

Schema	Key	Value
<code>org.gnome.desktop.interface</code>	<code>color-scheme</code>	<code>prefer-dark</code>
<code>org.gnome.desktop.interface</code>	<code>gtk-theme</code>	<code>Yaru-dark (override: GTK_THEME=)</code>
<code>org.gnome.desktop.interface</code>	<code>icon-theme</code>	<code>Yaru-dark</code>
<code>org.gnome.shell.ubuntu</code>	<code>color-scheme</code>	<code>prefer-dark</code>
<code>org.gnome.desktop.wm.preferences</code>	<code>theme</code>	<code>Yaru-dark</code>
<code>org.gnome.settings-daemon.plugins.color</code>	<code>night-light</code>	<code>disabled</code>

With `APPLY_GDM=1` (default), also sets GDM greeter dark via `sudo dbus-run-session gsettings ....`

apply-max-performance

Sets:

- Power plugin: no suspend on AC/battery, no idle dim, lid close does nothing
- Session idle delay: 0
- Screensaver: no blanking or lock on idle
- Night light: off
- `powerprofilesctl set performance` when available

How to verify

`gsettings get org.gnome.desktop.interface color-scheme`

`gsettings get org.gnome.desktop.interface gtk-theme`

`gsettings get org.gnome.settings-daemon.plugins.power sleep-inactive-ac-type`

`powerprofilesctl get`

Log out and back in to confirm GDM greeter if dark mode was applied.

How to customize

- Override themes: `GTK_THEME=Yaru-viridian bin/apply-dark-mode`
- Skip GDM: `APPLY_GDM=0 bin/apply-dark-mode`
- Revert individual keys with `gsettings reset ...` or GNOME Settings app
- Login/desktop asset mirrors: `Themes/login/, Themes/desktop/`

What rebuild does / does not do

Does	Does not
Install gdm3, gnome-shell, Yaru via apt	Change any gsettings Set performance power profile

Run both launchers after every fresh install (Chapter 3).

Chapter 8 — GNU Radio and Desktop SDR

What gets installed

Component	Version	Packages / path
GNU Radio	3.10.12.0	gnuradio, gnuradio-dev, gnuradio-doc
Companion blocks	apt	gr-osmosdr, gr-limesdr, gr-fosphor, gr-air-modes, gr-hpsdr, gr-dab, gr-satellites
SoapySDR	apt + Python	libsoapyhdr-dev, python3-soapyhdr, modules
Hardware libs	apt	RTL-SDR, HackRF, Airspy, bladeRF, Lime, UHD
Desktop apps	apt	gqrx-sdr, quisk, inspectrum, hacktv

SoapySDR modules on this host: HackRF, RTL-SDR (osmosdr), Airspy, bladeRF, Lime, MiriSDR, HydraSDR, PlutoSDR, Red Pitaya, remote, audio, UHD.

How it is installed

All packages in APT_SDR_HAM (rebuild Phase 3). Dev libraries in APT_CORE support building OOT modules.

Typical workflow:

```
grcc myflowgraph.grc      # compile Companion graph
gqrx                      # general receiver GUI
quisk                     # transceiver GUI
inspectrum capture.cf32   # visualize IQ files
```

For HackRF-specific host tools and URH, see Chapter 10.

How to verify

```
gnuradio-config-info --version
grcc --help | head -1
python3 -c "import gnuradio; print(gnuradio.__version__)"
python3 -c "import SoapySDR; print('SoapySDR OK!)"
SoapySDRUtil --info
gqrx --version 2>/dev/null || command -v gqrx
```

With hardware attached:

```
rtl_test -t              # RTL-SDR
hackrf_info              # HackRF
```

How to customize

- Add OOT modules: `sudo apt install gr-<name>` or build from source against `gnuradio-dev`
- GPU waterfall: `gr-fosphor` blocks in flowgraphs
- Filtered apt list: `apt-hamradio-dev-manifest.txt` (178 SDR/ham-related packages on full system)

What rebuild does / does not do

Does	Does not
Install full GNU Radio + gr-* stack	Calibrate specific SDR hardware
Install gqrx, quisk, inspectrum	Install SDRangel or SigDigger
Verify gnuradio-config-info, grcc, gqrx in verify_stack	Flash firmware on SDR devices

Chapter 9 — Ham Radio (Desktop)

What gets installed

Application	Command	Apt package	Role
fldigi	<code>fldigi</code>	<code>fldigi</code>	Digital modes (PSK, RTTY, ...)
WSJT-X	<code>wsjtx</code>	<code>wsjtx, wsjtx-data</code>	FT8, JT65, weak-signal
CHIRP	<code>chirpw, chirpc</code>	<code>chirp</code>	Radio programming
direwolf	<code>direwolf</code>	<code>direwolf</code>	Sound-card TNC / APRS
gpredict	<code>gpredict</code>	<code>gpredict</code>	Satellite pass prediction
grig	<code>grig</code>	<code>grig</code>	Hamlib rig control GUI
xastir	<code>xastir</code>	<code>xastir, xastir-data</code>	APRS map client
Hamlib	API	<code>libhamlib-dev, libhamlib-utils, python3-hamlib</code>	Rig control library

How it is installed

Apt packages in APT_SDR_HAM (rebuild Phase 3).

xastir debconf: rebuild preseeds `xastir/install-setuid` boolean false to avoid interactive install hangs.

```
fldigi &  
wsjtx &  
chirpw &  
direwolf -p  
gpredict &
```

```
grig &  
xastir &
```

How to verify

```
command -v fldigi wsjtx chirpw direwolf gpredict grig xastir  
python3 -c "import Hamlib; print('Hamlib OK')"  
bin/rebuild-machine --verify-only # checks fldigi, wsjtx, chirpw
```

Configure rig control in each app via Hamlib model selection.

How to customize

- Radio definitions: CHIRP stock configs + your radio CSV
- WSJT-X: ~/.config/WSJT-X/
- xastir maps: xastir-data package + user map sources
- direwolf: ~/.direwolf/direwolf.conf

What rebuild does / does not do

Does	Does not
Install all ham desktop apps + Hamlib	Configure radios or call signs
Preseed xastir setuid prompt	Set up APRS IS or igates
Verify fldigi, wsjtx, chirpw commands	Install fldigi/WSJT-X from source

Chapter 10 — HackRF and PortaPack Mayhem

What gets installed

Recommended firmware: PortaPack Mayhem v2.4.0

Apt packages

hackrf, hackrf-firmware, libhackrf-dev, hackrf-doc, inspectrum, hacktv, dfu-util, openocd, ARM GCC toolchain, plus GNU Radio/SoapySDR deps (Chapter 8).

Built from source (hackrf/build/)

Tool	Notes
hackrf_sweep	Spectrum sweep — not in older apt splits
hackrf_info, hackrf_transfer, ... libhackrf.so	Newer libhackrf (0.10.0) than apt alone Under hackrf/build/libhackrf/src/

Install prefix: `hackrf/local/` (`CMAKE_INSTALL_PREFIX`).

Release assets (`hackrf/releases/`)

File	Size	Purpose
FIRMWARE_mayhem_v2.4.0.zip	2.4 MB	USB flash bundle
COPY_TO_SDCARD_hackrf_mayhem_v2.4.0	201 MB	PortaPack microSD
no-world-map.zip		
OCI_hackrf_mayhem_v2.4.0.ppfw.tar	2.2 MB	Web flasher image

Extracted SD tree: `hackrf/sd-card/mayhem-v2.4.0/` (276 MB, 84 apps in APPS/)

Source repos (`hackrf/repos/`)

hackrf, mayhem-firmware (+ submodules), portapack-hackrf, urh, hacktv

Python venv

`hackrf/venv-urh/` — Universal Radio Hacker **2.10.0** (PyQt6). Launch: `bin/urh`

udev

`hackrf/scripts/99-hackrf.rules` installed to `/etc/udev/rules.d/` — plugdev access.

How it is installed

Automated (rebuild Phases 6–10):

1. Clone repos (shallow, skip if `.git` exists)

2. Init Mayhem submodules if needed
3. cmake + build HackRF host (skip: SKIP_HACKRF_BUILD=1)
4. hackrf/scripts/download-mayhem.sh
5. hackrf/scripts/prepare-sdcard.sh
6. URH venv if missing
7. hackrf/scripts/setup-udev.sh

Manual (hardware present):

```
source bin/hackrf-env
bin/hackrf-flash-mayhem           # extract flash bundle
bin/hackrf-prepare-sdcard        # re-extract SD payload
bin/hackrf-build-mayhem          # compile Mayhem from source (advanced)
bin/hackrf-download-mayhem       # re-fetch releases
```

PATH: source bin/hackrf-env adds hackrf/build/hackrf-tools/src and hackrf/local/bin.

How to verify

```
bin/rebuild-machine --verify-only
source bin/hackrf-env
hackrf_info           # needs USB device
hackrf/build/hackrf-tools/src/hackrf_sweep --help | head -1
bin/urh --version
ls hackrf/sd-card/mayhem-v2.4.0/APPS | wc -l
```

How to customize

- **Upgrade Mayhem:** Edit hackrf/scripts/download-mayhem.sh version URLs; re-run download + prepare
- **SD apps:** Copy subsets from sd-card/mayhem-v2.4.0/APPS/ to microSD
- **Full inventory:** hackrf/MANIFEST.txt

Mayhem v2.4.0 highlights

On-device: Morse RX/TX, RTTY, FPV detect, ADSB, ACARS, BLE, TPMS, KeeLoq, EPIRB, SAME, MDC-1200, P25, KISS TNC, Looking Glass, SubGHz, Flipper TX, waterfall designer, time sink.

SD apps include: fpv_detect, kiss_tnc, keeloqtx, siggen, fmradio, sstvrx, wardrivemap, waterfall_designer, and more.

What rebuild does / does not do

Does	Does not
Clone repos, build tools, download Mayhem	DFU-flash firmware to hardware
Create URH venv, install udev	Format or write microSD in a reader
Verify zip, SD tree, hackrf_sweep	Test with HackRF USB attached

Chapter 11 — Flatpak Applications

What gets installed

App	Flatpak ID	Version (this host)
Telegram	org.telegram.desktop	6.9.3

Runtime dependency: flatpak package from APT_CORE.

How it is installed

Rebuild Phase 4:

```
flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpakrepo
flatpak install -y flathub org.telegram.desktop
```

Skip with SKIP_TELEGRAM=1 bin/rebuild-machine.

How to verify

```
flatpak list --app | grep telegram
flatpak run org.telegram.desktop --version 2>/dev/null || true
bin/rebuild-machine --verify-only
```

How to customize

```
flatpak update org.telegram.desktop
flatpak override --user org.telegram.desktop ... # permissions, env
bin/apply-dark-mode sets prefer-dark color scheme; Flatpak GTK4
apps pick this up via portal when supported.
```

What rebuild does / does not do

Does	Does not
Install flatpak via apt Add flathub remote + Telegram Treat Telegram miss as non-fatal on install (warns in log)	Install SDRangel, SigDigger (not on Flathub) Pin Telegram to a specific commit Install other Flatpak apps by default

Chapter 12 — Machine Utilities

What gets installed

Workspace scripts for inventory, storage benchmark, and GPU stress — no dedicated apt packages beyond shared GPU utils (mesa-utils, vulkan-tools).

Utility	Launcher	Script	Output
Dell inventory	bin/dellmerge	scripts/dell/dellmerge.sh	dellmerge.sh *.report files
Storage survey	bin/iotest	scripts/storage/iotest.sh	iotest.sh (sudo)
GPU stress	bin/gpu-stress	scripts/gpu/gpu-stress.sh	Vulkan/EGL per GPU
EFI / BIOS timing	bin/efi-timing-suite	scripts/efi/efi-timing-suite.sh	B1GMB42.timing (sudo)

Example reports in workspace: blgmb42.report, B1GMB42.ioperf, B1GMB42.timing (from prior runs).

How it is installed

Scripts ship with the workspace. Rebuild Phase 10 runs `chmod +x` on `bin/*` and `scripts/*/*.sh`.

```
bin/dellmerge > blgmb42.report
sudo bin/iotest
bin/gpu-stress 60 vkcube
sudo bin/efi-timing-suite
```

before/after BIOS A/B changes

How to verify

```
[[ -x bin/dellmerge && -x bin/iotest && -x bin/gpu-stress && -x bin/efi-timing-suite  
bin/rebuild-machine --verify-only # checks dellmerge, gpu-stress, iotest, apply  
head -20 blgmb42.report 2>/dev/null || bin/dellmerge | head -20
```

How to customize

- Edit scripts/dell/dellmerge.sh to add inventory fields
- gpu-stress accepts duration and backend (vkcube default)
- iotest targets block devices — read script header before running on production pools
- efi-timing-suite writes a machine-local timing baseline; re-run after BIOS changes for A/B compare

What rebuild does / does not do

Does	Does not
chmod utility launchers	Run dellmerge, iotest, or efi-timing automatically
Verify launcher executables exist	Archive reports to a fixed path

Chapter 13 — FactoryDocs (Workspace Archive)

What gets installed

Nothing on the Linux host automatically. FactoryDocs is a sorted archive of Dell T5810 vendor support packages for Windows recovery, firmware, and reference — stored only in the workspace.

Metric	Pre-crash	Current
Sorted packages	101	19
GPU drivers (FirePro/Quadro)	Yes	Missing
PERC H710 driver/firmware	Yes	Missing
Audio / input drivers	Yes	Missing
Win7/10/WinPE CAB packs	Yes	Missing

Full pre-crash file list: FactoryDocs/MANIFEST-pre-crash.txt (91 items flagged MISS).

Layout

Folder	Contents
System-T5810/	BIOS, chipset, ME, TPM, manuals
GPU/	AMD FirePro, NVIDIA (empty — re-download)
Storage/	PERC H710, Intel RST, SSD/HDD firmware
Network/	Intel Ethernet
Dell-Management/	Command Update, Configure
Expansion-Cards/	Serial, Thunderbolt docs
_Misc/	Non-Dell packages
_incoming/	Drop new Dell downloads here

How it is installed

Ingest new downloads:

```
cp ~/Downloads/* ~/Documents/IndianaDell/FactoryDocs/_incoming/  
python3 ~/Documents/IndianaDell/FactoryDocs/_sort_factory_docs.py
```

Priority re-downloads from Dell T5810 drivers:

1. GPU/AMD-FirePro/Windows/ — Video_Driver_C5FPW (W5000/W5100)
2. Storage/RAID-Controller-PERC/Windows/ — PERC H710
3. System-T5810/Windows-10/ — T5810-win10 CAB
4. Audio/Windows/ — Audio_Driver_5P33P
5. Dell-Management/Windows/ — Command Configure, Monitor

Windows installs use Dell CAB/EXE packages from these folders — not apt.

How to verify

```
find FactoryDocs -type f ! -path '*/_incoming/*' | wc -l  
cat FactoryDocs/README.md  
grep MISS FactoryDocs/MANIFEST-pre-crash.txt | wc -l
```

How to customize

- `_sort_factory_docs.py` dedupes and sorts by hardware category
- Cross-reference hardware manual: `B1GMB42-slot-port-inventory.md` for what hardware needs drivers

What rebuild does / does not do

Does	Does not
Nothing	Copy FactoryDocs to system paths Install Windows drivers Download missing CABs

FactoryDocs recovery is a **manual** ongoing task (Chapter 3 item 7).

Chapter 14 — Gaps and Limits

Documented boundaries of what IndianaDell does **not** install or support on this host.

Not installed

Item	Notes
SDRangel, SigDigger	Not on Flathub; use gqrx + URH + inspectrum
Rust SDR crate workspace	Toolchain only — add crates per project with cargo add
ZFS / disk layout tools	Out of scope — handled at OS install time
Windows / dual-boot	FactoryDocs holds CABs; no auto-install
HackRF hardware test	No device attached at last verify (2026-07-05)
Ventoy seed on every boot	Manual — run ~/bin/seed-ventoy-persistence.sh after changes

Lost in TPM/ZFS crash

Item	Recovery
Pre-crash full apt list	Not found — use apt-full-manifest.txt from current rebuild
Pre-crash FactoryDocs	82/101 packages missing — see MANIFEST-pre-crash.txt
Pre-crash apt-hamradio list	Regenerated by rebuild save_manifests()

GPU / ROCm matrix

Capability	W5000 / W5100 on Ubuntu 26.04
Desktop amdgpu	Yes — with etc/ configs
Vulkan (vkcube)	Yes
OpenCL (clinfo)	Yes (limited)
ROCm HIP / ML training	No — not in AMD ROCm support matrix for FirePro W5000/W5100

Use `bin/amd-preflight` before `bin/amd-install`; expect warnings for these GPUs.

Rebuild intentional omissions

These remain **manual** by design (see Chapter 3):

- `sudo bin/apply-amdgpu`
- `bin/apply-dark-mode`, `bin/apply-max-performance`
- `bin/themes-extract`, `sudo bin/themes-install-boot`
- `bin/hackrf-flash-mayhem` (DFU with hardware)
- `bin/amd-install` (optional ROCm)

Future enhancement could fold GNOME/theme steps into rebuild; current manual documents the gap explicitly.

Encryption / TPM

Hardware manual recommends **not** re-enabling ZFS encryption until TPM + recovery strategy is documented. IndianaDell rebuild does not touch encryption.

ZFS boot import

The installed host **must** keep force import enabled:

```
# /etc/default/zfs
ZPOOL_IMPORT_OPTS="-f"
```

`bin/rebuild-machine` does **not** write this file. After reinstall or recovery, verify it yourself (Chapter 3 checklist). Without `-f`, boot can hang when pools need force-import (post-export hostid mismatch, unclean shutdown).

When something fails

1. Read `scripts/rebuild/last-run.log`

2. Run `bin/rebuild-machine --verify-only` for targeted MISS lines
3. Re-run individual phases (apt, HackRF build, Mayhem download) manually
4. Check chapter-specific verify sections

Reporting issues

```
Capture:
bin/dellmerge > debug.report
bin/rebuild-machine --verify-only 2>&1 | tee verify.log
uname -a && lsb_release -a
```

Chapter 15 — Ventoy Live Session & Persistence

Portable Ubuntu 26.04 on the **Wiggly** Ventoy stick, with a writable overlay so login state, apps, Grok, and IndianaDell survive reboots.

What gets persisted

Item	Location (live boot)	Seeded to casper image
User home	/home/ubuntu	cow/upper/home/ubuntu/
Installed packages	dpkg overlay	cow/upper/var/lib/dpkg/
GDM autologin	/etc/gdm3/custom.conf	cow/upper/etc/gdm3/
Grok auth + sessions	~/.grok/	same (never in git)
GitHub CLI auth	~/.config/gh/	same
SSH keys	~/.ssh/	same
Runtime source	/home/user/ when ZFS rpool is available	pulled at login via <code>resolve-secrets.sh</code>
IndianaDell workspace	/Documents/IndianaDell	same (git clone or rsync)

Item	Location (live boot)	Seeded to casper image
PATH over- rides	~/.config/indianadell/path.sh	path.sh

Persistence image: /persistence/ubuntu-26.04.dat (24 GB ext4, label casper-rw) on the internal Ventoy exFAT volume (**Wiggly**, sdc1).

Ventoy config: ventoy/ventoy.json maps ubuntu-26.04-desktop-amd64.iso to that .dat file with autoselect: 1. Canonical copy in scripts/ventoy/ventoy.json.

How it is installed

One-time setup from Tower5810 (Wiggly mounted at /mnt/wiggly):

```
sudo mount -o uid=$(id -u),gid=$(id -g) /dev/disk/by-label/Wiggly /mnt/wiggly
bin/setup-wiggly-ventoy # verify ISO, ventoy.json, .dat filesystem
```

Extend an undersized image: sudo scripts/ventoy/ExtendPersistentImg.sh /mnt/wiggly/persistence/ubuntu-26.04.dat <MB> then resize2fs on the loop device if needed.

Seed session state from a running session with Wiggly mounted (e.g. /mnt/wiggly):

```
# One-time or after changes – seeds current ubuntu session into the .dat image
~/bin/seed-ventoy-persistence.sh
# or, if the image is already mounted:
PERSIST_MOUNT=/mnt/persist-check ~/bin/seed-ventoy-persistence.sh
```

The seed script copies home, dpkg/apt state, GDM autologin, SSH keys (including /home/user/.ssh/id_rsa when present), and the IndianaDell tree.

Login experience (configured)

1. **GDM autologin** — user ubuntu (/etc/gdm3/custom.conf)
2. **PATH** — IndianaDell bin/ and scripts/ override system (~/.config/indianadell/path.sh)
3. **Grok autostart** — Ptyxis fullscreen, resumes IndianaDell session (~/.config/autostart/grok-indianadell.desktop)

Launcher: ~/bin/grok-indianadell-launch.sh

resolve-secrets.sh materializes secrets from /home/user when rpool exists, else uses Ventoy \$HOME.

Runs ~/bin/seed-ventoy-persistence.sh **before** Grok (logs to ~/.cache/seed-ventoy.log).

Seed verifies **internet + DNS** first; if down, offers NetworkManager

bring-up or skip. Default session: ~/Documents/IndianaDell (session ID in script env vars).

ZFS recovery (rpool + bpool)

Manual: docs/B1GMB42-zfs-recovery.md + B1GMB42-zfs-recovery.pdf
(repo root and DOSBOOT/IndianaDell/recovery/).

Boot Ventoy Ubuntu live — **do not** use the broken installed system as root.

```
sudo apt-get install -y zfsutils-linux
cd ~/Documents/IndianaDell # or /media/.../DOSBOOT1/IndianaDell/recovery
sudo ./mount-rpool-recovery.sh mount
sudo ./scripts/recovery/mount-bpool-recovery.sh mount
sudo ./mount-rpool-recovery.sh chroot
# repair inside chroot; then exit and umount both scripts
```

Before rebooting the installed system: ensure /etc/default/zfs has ZPOOL_IMPORT_OPTS="-f". Recovery scripts pass -f on import; the host boot path needs this default or boot can hang after export/unclean shutdown. Kernel one-shot: zfsforce=1.

No IndianaDell? Same manual, Section 3 — raw zpool import commands.

Deploy kit to DOSBOOT: bin/deploy-dosboot-recovery (from Tower5810).

GitHub repository

Full workspace (including FactoryDocs): <https://github.com/webaugur/IndianaDell> (private)

```
bin/pull-repo --verify # IndianaDell + hackrf/repos + LFS + stack verify
bin/push-repo # push main (SSH default)
```

HTTPS push (optional): INDIANADELL_REMOTE=<https://github.com/webaugur/IndianaDell.git> after gh auth login.

Large FactoryDocs installers (>100 MB) use **Git LFS**. bin/pull-repo runs git lfs pull.

How to verify

Boot Ventoy ▢ Ubuntu 26.04 (persistence auto-selected). Then:

```
findmnt / | grep -q cow && echo "persistence overlay active"
grep AutomaticLogin=ubuntu /etc/gdm3/custom.conf
echo "$INDIANADELL_ROOT" # should be ~/Documents/IndianaDell
which dellmerge pull-repo push-repo grok
```

```
bin/pull-repo --verify
google-chrome --version
```

How to customize

Goal	Action
Re-seed after changes	~/bin/seed-ventoy-persistence.sh
Change Grok session	Edit GROK_SESSION_ID in grok-indianadell-launch.sh
Disable autostart	Remove ~/.config/autostart/grok-indianadell.desktop, re-seed
Enlarge persistence	scripts/ventoy/ExtendPersistentImg.sh (+ resize2fs if needed)
Verify/fix Ventoy layout	bin/setup-wiggly-ventoy from Tower5810

What rebuild does / does not do

Does	Does not
Install Chrome, gh, git-lfs when run on live session	Configure Ventoy ventoy.json
Document seed script in this chapter	Auto-run seed on reboot
	Manage Ventoy ISO partition layout

Appendix A — bin/ Launchers

All launchers live in ~/Documents/IndianaDell/bin/. **PATH** is set automatically via ~/.config/indianadell/path.sh (IndianaDell tools override system binaries).

Launcher	Runs	Chapter
rebuild-machine	scripts/rebuild-machine.sh	2
build-software-manual	scripts/docs/build-software-manual.sh	1
build-all-docs	scripts/docs/build-all-docs.sh	1

Launcher	Runs	Chapter
pull-repo	scripts/github/pull-all.sh — IndianaDell + nested repos + LFS	
push-repo	bin/push-repo □ GitHub	15
setup-wiggly-ventoy	webau- gur/IndianaDell (SSH default) scripts/ventoy/setup-wiggly-ventoy.sh — ISO + ventoy.json + .dat	15
setup-perc-ventoy	scripts/perc/setup-perc-ventoy.sh — H710 FreeDOS/IT kit on Wiggly	hardware / PERC doc
build-zfs-recovery-doc	scripts/docs/build-zfs-recovery-doc.sh	2, 105
build-trifold-slick	docs/sales/B1GMB42-trifold.html □ sales PDFs	
deploy-dosboot-recovery	scripts/recovery/deploy-to-dosboot.sh	2, 105
efi-timing-suite	scripts/efi/efi-timing-suite.sh	6, 12
dellmerge	scripts/dell/dellmerge.sh	12
gpu-stress	scripts/gpu/gpu-stress.sh	6, 12
iotest	scripts/storage/iotest.sh	12
apply-amdgpu	etc/apply.sh	6
amd-install	amd-radeon/install-all.sh	6
amd-preflight	amd-radeon/00-preflight.sh	6

Launcher	Runs	Chapter
amd-verify	amd-radeon/04-verify.sh	6
amd-uninstall	amd-radeon/uninstall.sh	6
apply-dark-mode	scripts/gnome/apply-dark-mode.sh	5
apply-max-performance	scripts/gnome/apply-max-performance.sh	5
themes-extract	Themes/scripts/extract-all.sh	5
themes-install-boot	Themes/scripts/install-boot-theme.sh	5
themes-restore-boot	Themes/scripts/install-boot-theme.sh --restore-stock	5
hackrf-env	sources	10
urh	hackrf/scripts/env.sh hackrf/scripts/launch-urh.sh	10
hackrf-setup-udev	hackrf/scripts/setup-udev.sh	10
hackrf-download-mayhem	hackrf/scripts/download-mayhem.sh	10
hackrf-prepare-sdcard	hackrf/scripts/prepare-sdcard.sh	10
hackrf-flash-mayhem	hackrf/scripts/flash-mayhem.sh	10
hackrf-build-mayhem	hackrf/scripts/build-mayhem.sh	10

Ventoy session (scripts/ventoy/ □ ~/bin via install-ventoy-session.sh):

Script	Purpose
seed-ventoy-persistence.sh	Snapshot session into Ventoy casper image
seed-network-check.sh	Internet/DNS check before seed
grok-indianadell-launch.sh	Seed then Grok fullscreen autostart
install-ventoy-session.sh	Install helpers, autostart, PATH

Script	Purpose
mount-rpool-recovery.sh	ZFS rpool chroot recovery (workspace root)
mount-bpool-recovery.sh	ZFS bpool mount at /recovery/boot (scripts/recovery/)

Note: hackrf-env must be **sourced**, not executed: source bin/hackrf-env

Sudo required: apply-amdgpu, themes-install-boot, themes-restore-boot, iotest, hackrf-setup-udev (udev install), amd-install, efi-timing-suite, mount-rpool-recovery.sh, mount-bpool-recovery.sh

Appendix B — Apt Packages by Chapter

Source of truth: scripts/rebuild/package-lists.sh (APT_CORE + APT_SDR_HAM).

Total: 91 packages installed by bin/rebuild-machine (38 core + 53 SDR/ham).

Full system snapshot: apt-full-manifest.txt (~2257 packages after rebuild).

SDR/ham filter: apt-hamradio-dev-manifest.txt (~178 related packages).

Packages below are grouped by manual chapter. Shared dev libraries appear under Chapter 4 and are reused by Chapters 8-10.

Chapter 4 — Development

build-essential, cmake, pkg-config, git, curl, wget, unzip, python3-pip, python3-venv, python3-dev, python3-numpy, python3-scipy, python3-matplotlib, python3-yaml, python3-requests, python3-pyqt5, python3-psutil, libssl-dev, clang, llvm-dev, libclang-dev, libusb-1.0-0-dev, libfftw3-dev, libvolk-dev, portaudio19-dev, libsndfile1-dev, libboost-dev, libboost-program-options-dev, pandoc, texlive-latex-recommended, texlive-fonts-recommended, texlive-xetex, gh

Chapter 6 — GPU and Display

vulkan-tools, mesa-utils, mesa-utils-bin, clinfo

Chapter 8 — GNU Radio and SDR

gnuradio, gnuradio-dev, gnuradio-doc, gr-osmosdr, gr-limesdr, gr-fosphor, gr-air-modes, gr-hpsdr, gr-dab, gr-satellites, libsoapysdr-dev, python3-soapysdr, soapysdr-module-osmosdr, soapysdr-module-mirisdr, uhd-soapysdr, rtl-sdr, librtlsdr-dev, airspy, libairspy-dev, bladerf, libbladerf-dev, limesuite, limesuite-udev, uhd-host, libuhd-dev, gqrx-sdr, quisk, inspectrum, hacktv

Chapter 9 — Ham Radio

libhamlib-dev, libhamlib-utils, python3-hamlib, fldigi, wsjtx, wsjtx-data, chirp, direwolf, gpredict, grig, xastir, xastir-data

Chapter 10 — HackRF and Mayhem

hackrf, hackrf-firmware, libhackrf-dev, hackrf-doc, dfu-util, openocd, gcc-arm-none-eabi, binutils-arm-none-eabi, libnewlib-arm-none-eabi, ccache, lz4, bzip2

(Also uses Chapter 8 packages for GNU Radio/SoapySDR integration.)

Chapter 11 — Flatpak

flatpak (application org.telegram.desktop installed via flatpak, not apt)

Chapters 5, 7, 12, 13 — No dedicated apt arrays

Themes, GNOME prefs, machine utilities, and FactoryDocs use workspace scripts or Ubuntu desktop packages already on the base install (gdm3, gnome-shell, plymouth, Yaru themes) — not enumerated separately in package-lists.sh.